

Magneto-optical imaging non-destructive testing image fusion of welding defects under controlled rotating magnetic field excitation based on the BM3D-DLIA method

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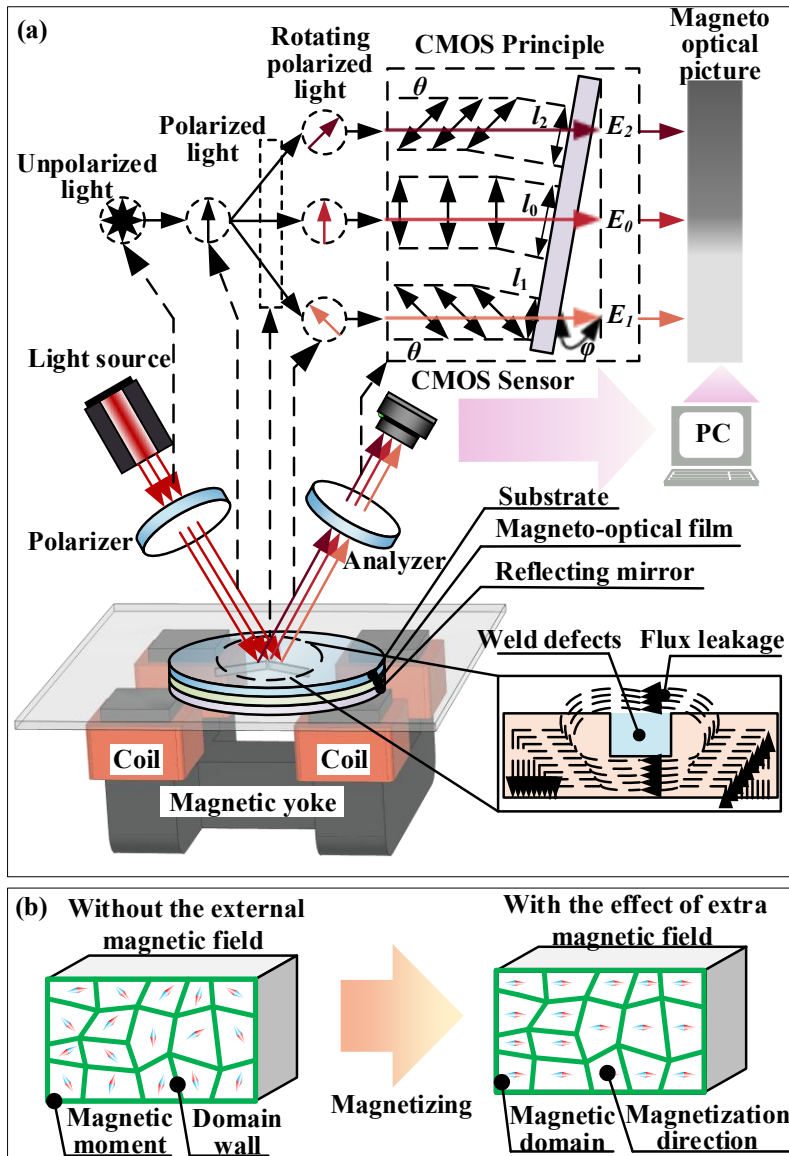
Background

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Magneto-optical image noise reduction

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Magneto-optical image fusion

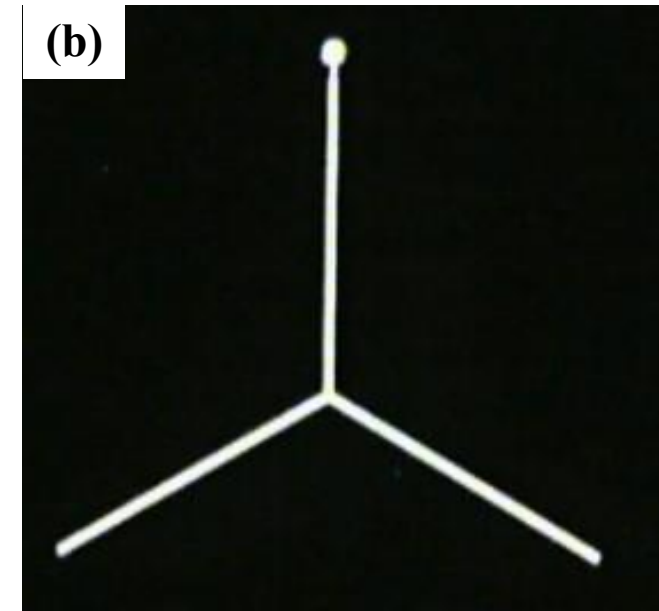
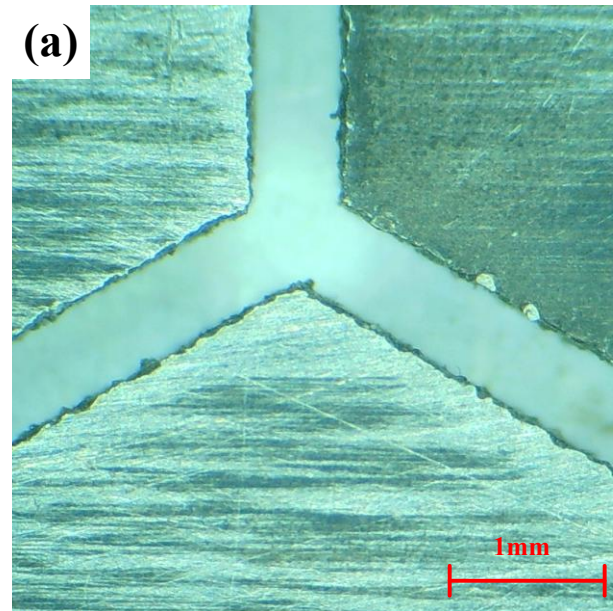
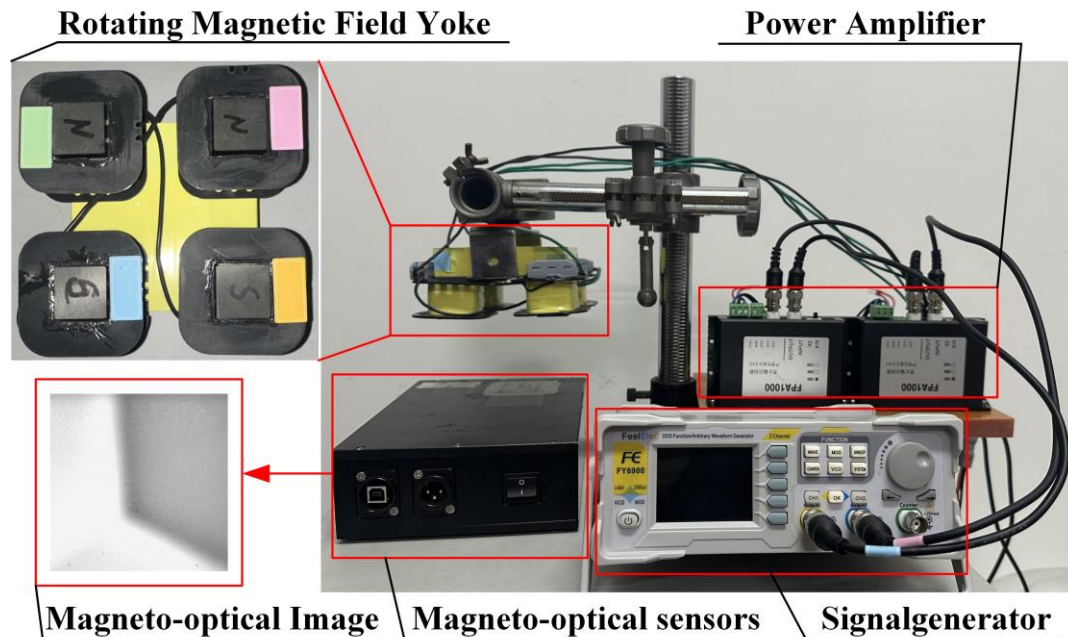


- Natural light is converted into linearly polarized light by the polarizer and irradiates the magneto-optic thin film.
- Under the influence of defect leakage magnetic fields, the film exhibits the magneto-optic effect, causing the polarization plane of the reflected light to rotate.
- This rotation angle is converted into an intensity signal by the analyzer.
- CMOS captures and generates high-resolution defect images.

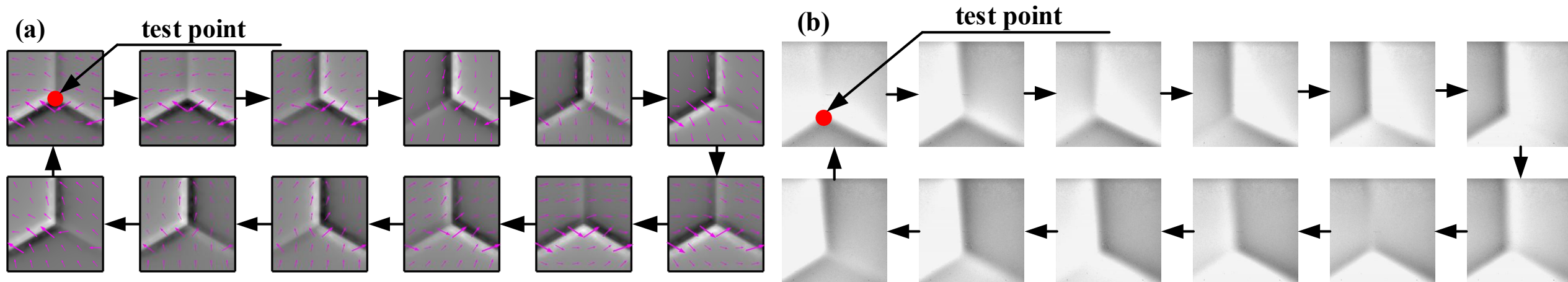
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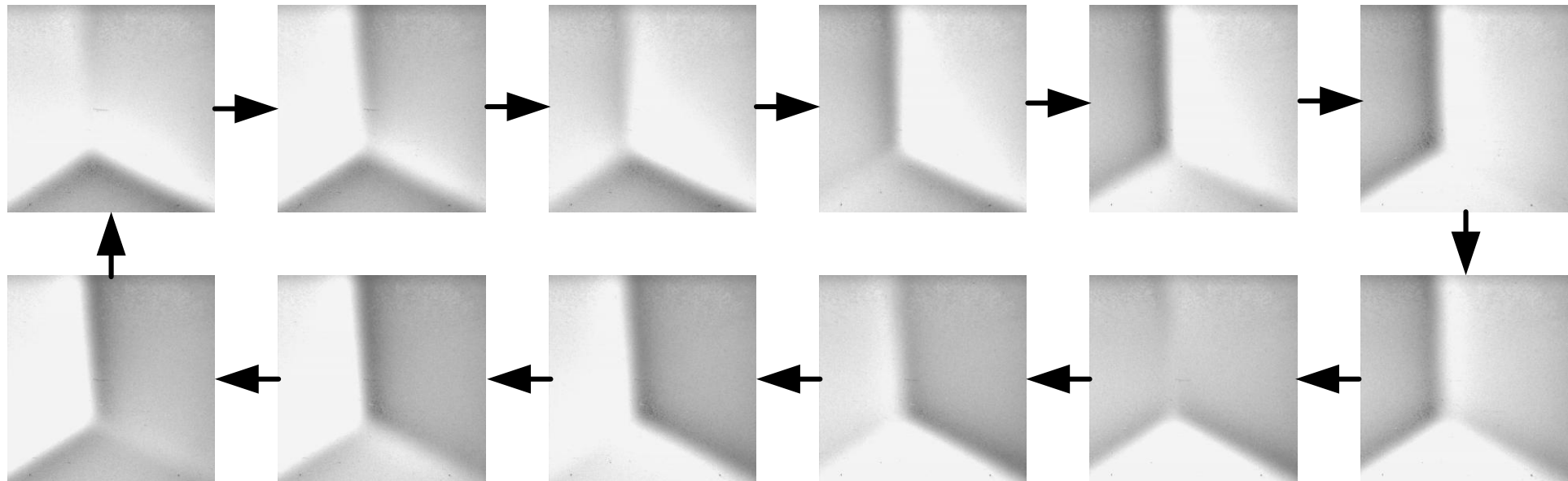
To simulate anisotropic defects encountered in real-world welding defect detection, a trident-shaped defect as shown in the figure was employed for simulation and experimental analysis of magnetic optical imaging of welding defects under controlled rotating magnetic field excitation.



- The comparison between finite element simulation results (Fig.a) and actual experimental data (Fig.b) demonstrates consistency in magnetic field distribution, **defect response signals, and imaging characteristics.**
- The simulated defect exhibits anisotropic properties. As the magnetic field direction changes, **the variation in leakage magnetic field intensity becomes clearly discernible** when the magnetic field is perpendicular to the anisotropic defect. This allows for distinct observation of leakage magnetic field fluctuations at different angles, **revealing the morphology of the weld defect.**



- Multi-angle magnetic-optical images of welding defects obtained from a controllable rotating magnetic field are fused by integrating magnetic-optical properties.
- This approach enhances defect detection accuracy and improves the quality of magnetic-optical image fusion for welding defects.



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Magneto-optical image noise reduction

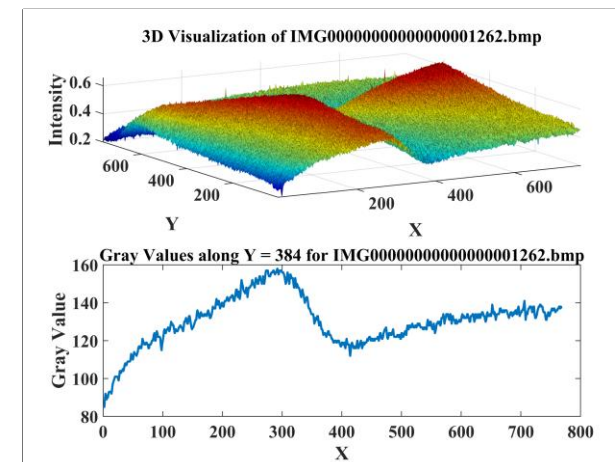
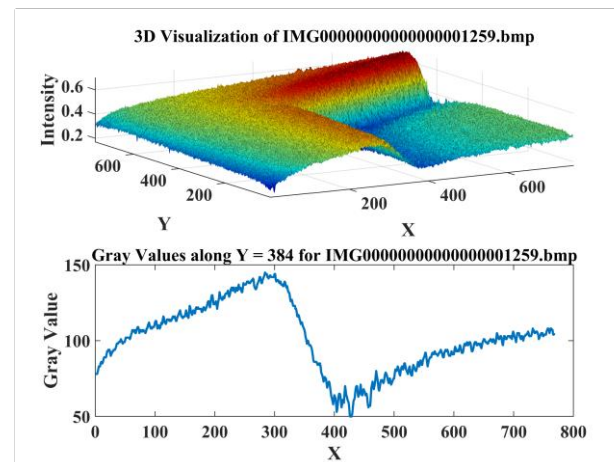
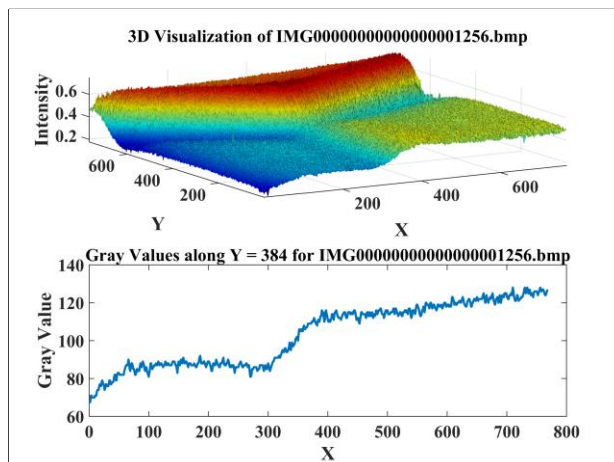
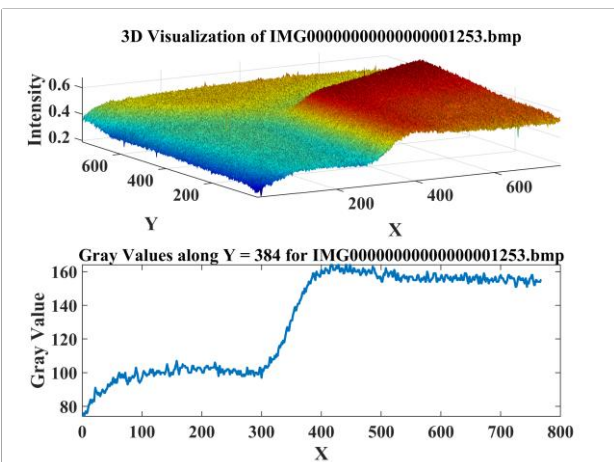
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Magneto-optical image fusion

Magneto-optical image noise reduction

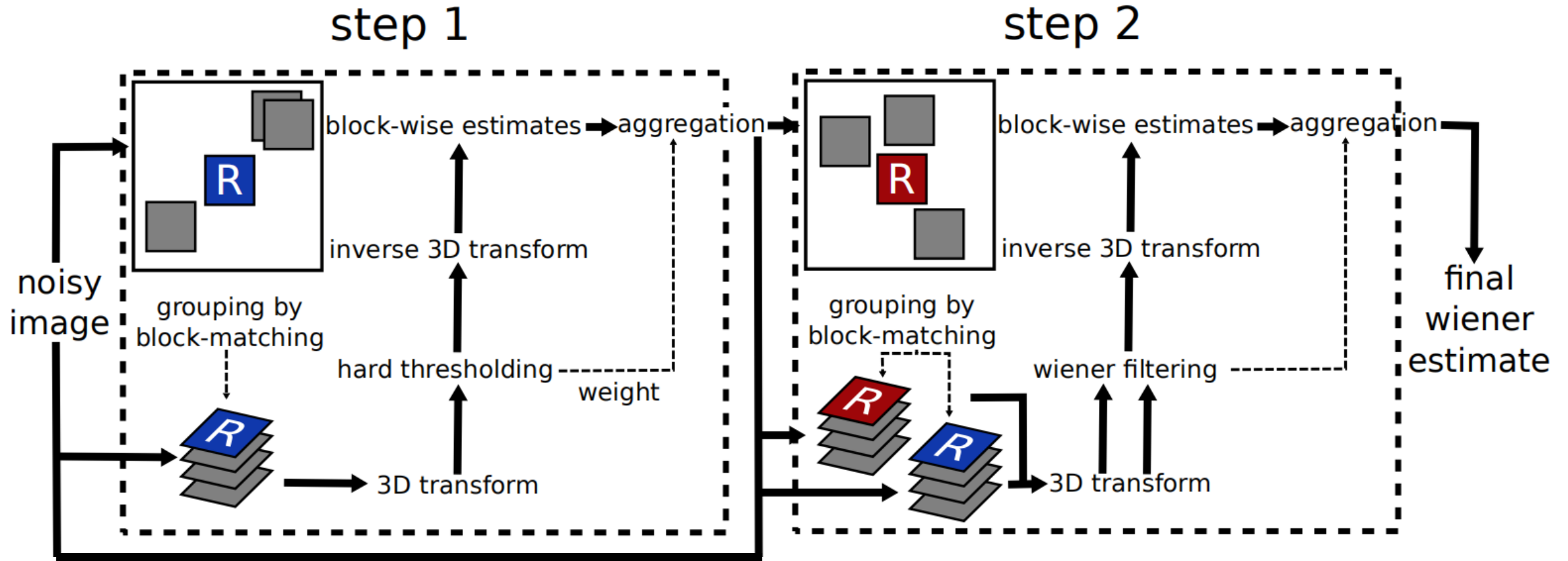
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- Changes in the direction of the rotating magnetic field cause significant variations in the gray-scale values of defect magneto-optical images.
- Low-frequency AC signal excitation sources induce irregular textured regions in magneto-optical images.
- Reducing speckle noise in magneto-optical images mitigates its impact on subsequent magneto-optical image fusion.



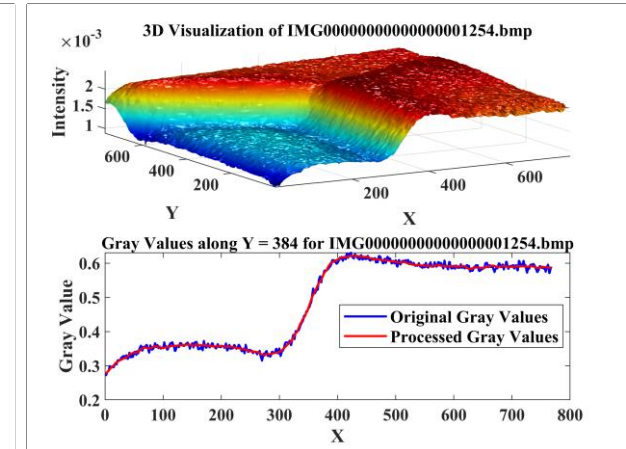
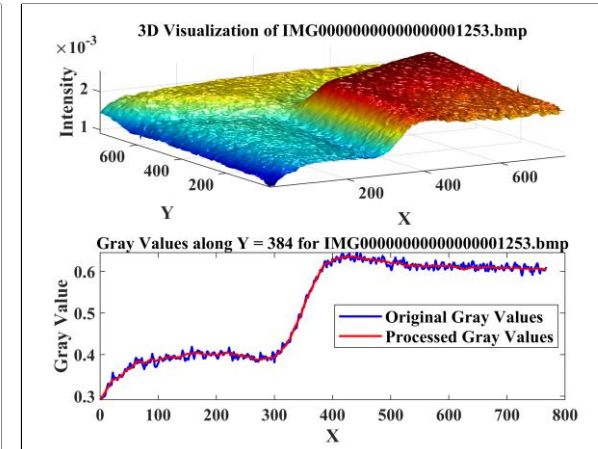
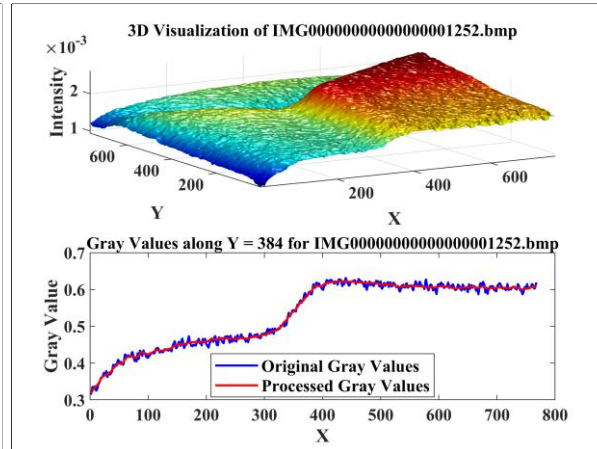
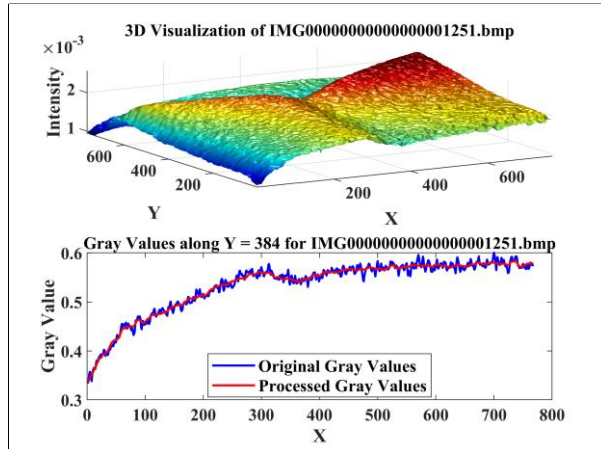
Magneto-optical image noise reduction

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BM3D (Block-Matching and 3D Filtering)

- After BM3D processing, noise is significantly reduced and the intensity distribution becomes smoother.
- The fluctuations in the original grayscale curve diminish after BM3D processing, resulting in a more even amplitude.
- The trend of the grayscale curve remains unchanged, indicating that BM3D preserves key features while enhancing image consistency.



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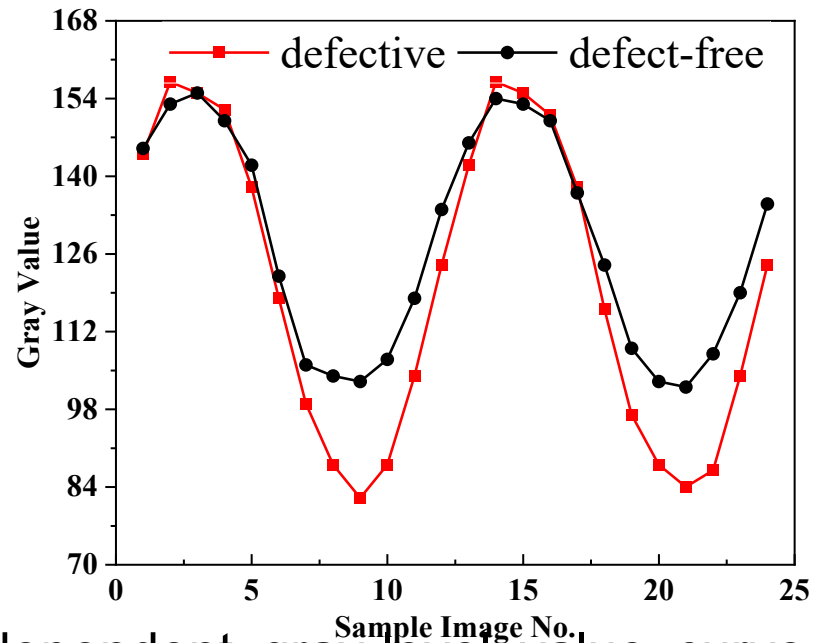
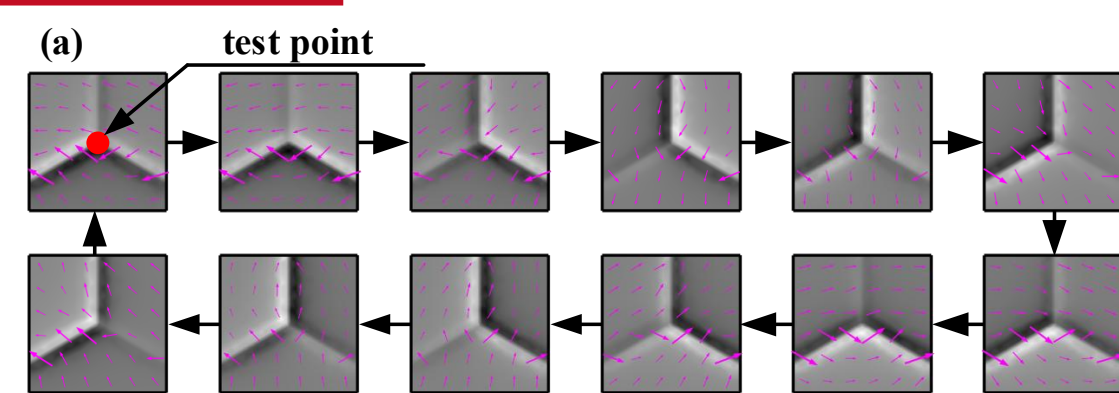
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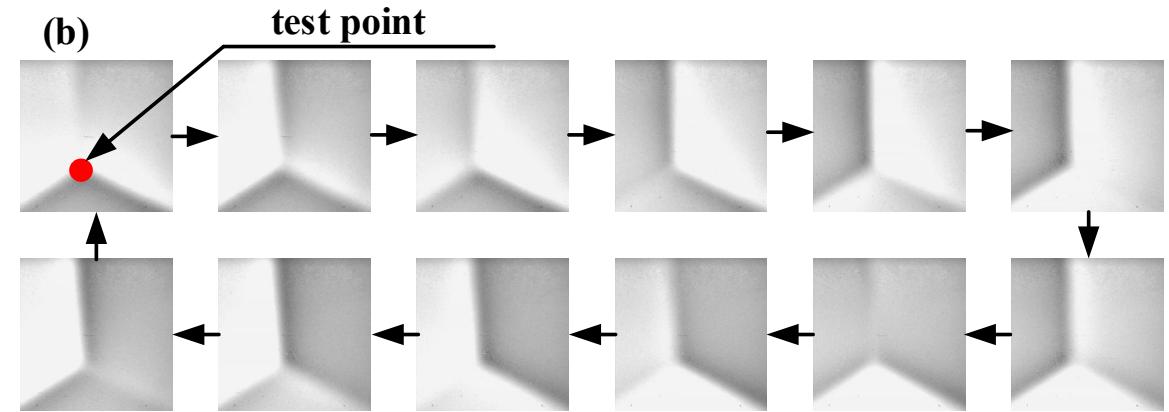
Magneto-optical image noise reduction

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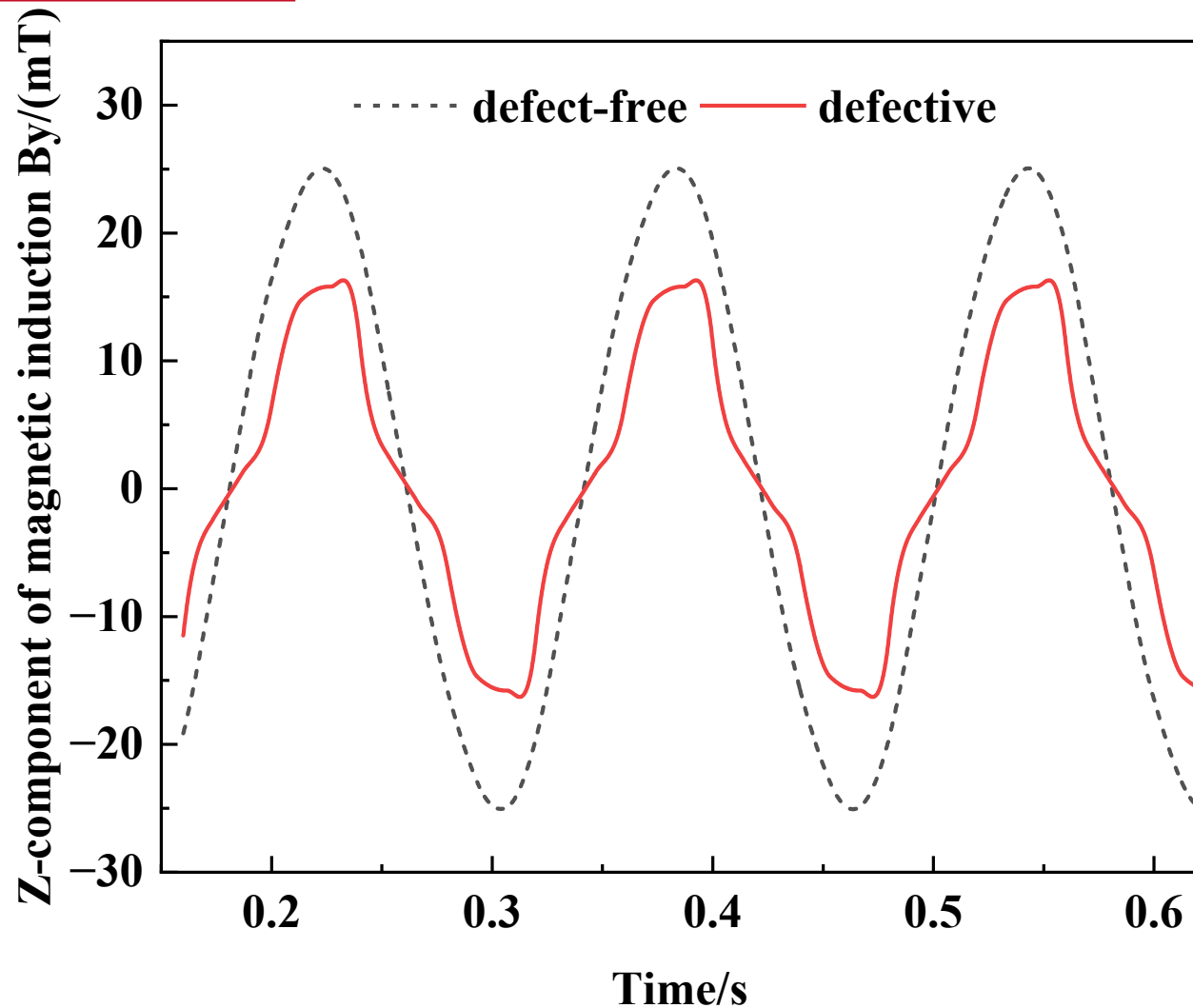
Magneto-optical image fusion



Time-dependent gray-level value curves of two test points with and without defects in magneto-optical images



- The gray values at both defect-free and defective locations exhibit trigonometric-like fluctuations.
- When defects are present, the curve displays higher gray values at peaks and troughs.
- The fluctuation cycles of the defect-free and defective curves are essentially consistent.

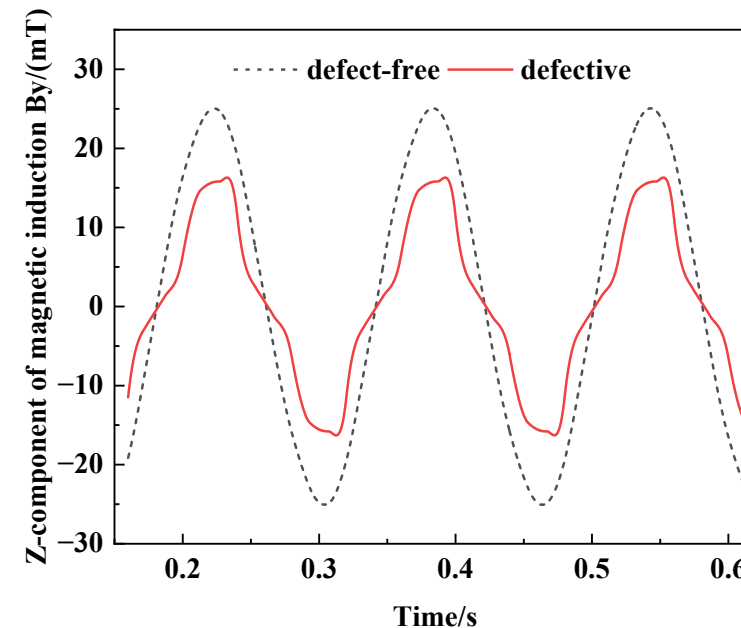
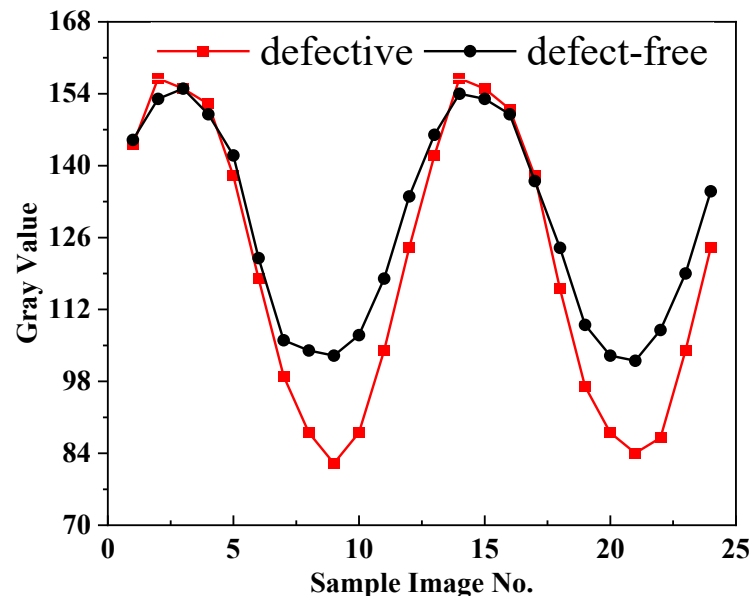


- At present, the Z-component of the magnetic flux density at this point exhibits a trigonometric variation trend with the excitation signal.
- When defects are present, the curve becomes distorted.

Time-dependent curve of the Z-component of magnetic flux density at test points with and without defects

- Each pixel in a magneto-optical image forms a time series of grayscale values across consecutive images.
- Periodic excitation by a rotating magnetic field causes defect-related signals within the magneto-optical response to exhibit this periodic characteristic.

How to Achieve Fusion of Multi-Angle Magneto-Optical Images Using Periodic Features?



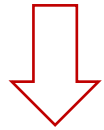
**Reference
signal**

$$\begin{cases} R_X(t) = \sin(\omega_r t) & \text{In-phase Reference, } X\text{-channel} \\ R_Y(t) = \cos(\omega_r t) & \text{Quadrature Reference, } Y\text{-channel} \end{cases}$$



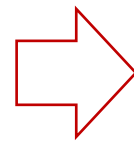
**Indexing for
Image
Discretization**

$$\begin{cases} ref_sin = \sin\left(2 \cdot \pi \cdot numRefCycles \cdot \frac{n-1}{n}\right) \\ ref_cos = \cos\left(2 \cdot \pi \cdot numRefCycles \cdot \frac{n-1}{n}\right) \end{cases}$$



**Multiply point-
by-point with
each of the two
reference
signals**

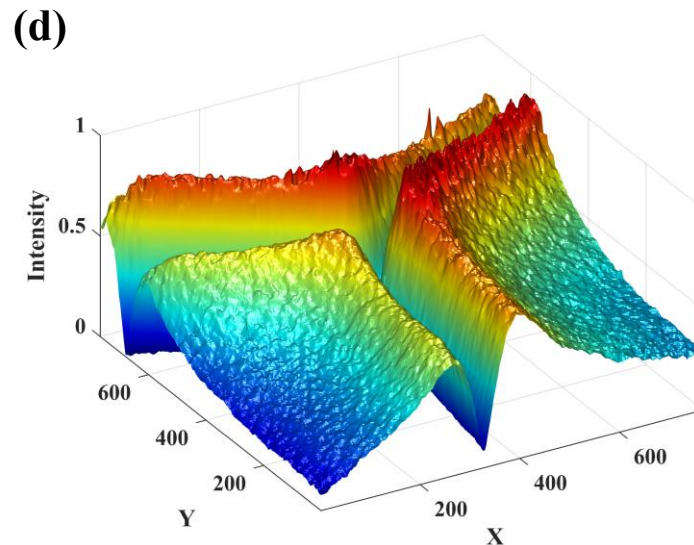
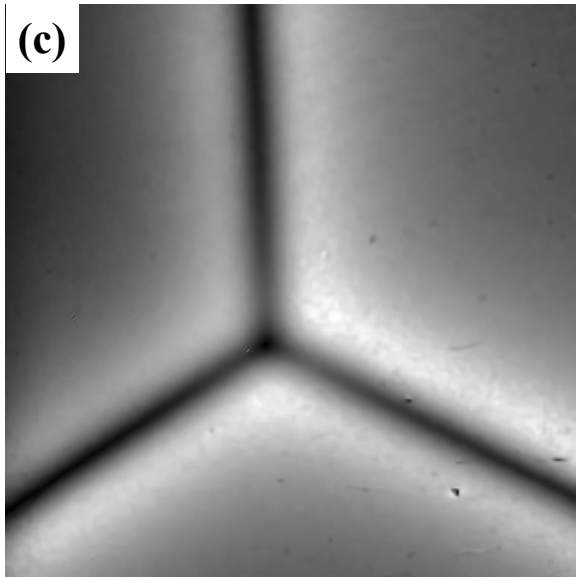
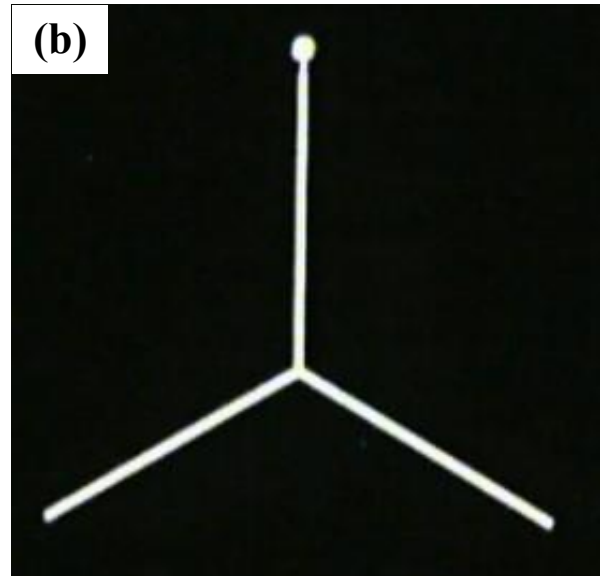
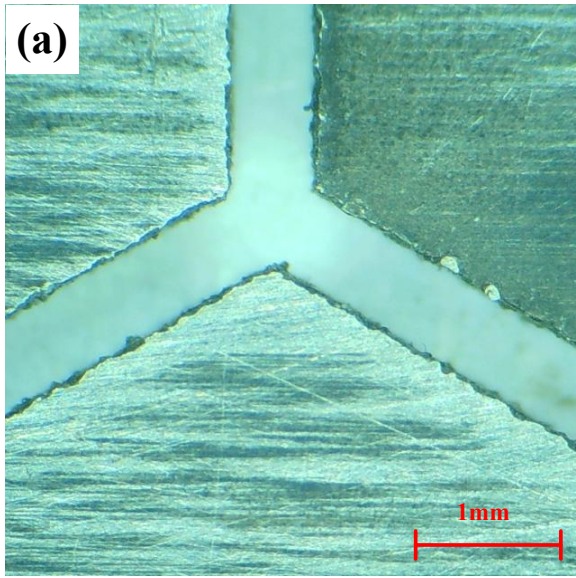
$$\begin{cases} P_X[k] = S_{pixel}[k] \cdot R_X[k] \\ P_Y[k] = S_{pixel}[k] \cdot R_Y[k] \end{cases}$$



$$\begin{cases} P_X(t) = 2A [\cos(\phi_s) - \cos(2\omega_r t + \phi_s)] \\ P_Y(t) = 2A [\sin(2\omega_r t + \phi_s) + \sin(\phi_s)] \end{cases}$$

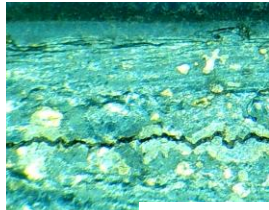
Low-pass filter $\begin{cases} X = LPF\{P_X[k]\} \approx 2A\cos(\phi_s) \\ Y = LPF\{P_Y[k]\} \approx 2A\sin(\phi_s) \end{cases}$

Fusion Image $\begin{cases} R = \sqrt{X^2 + Y^2} = \sqrt{\left(\frac{A}{2}\cos\phi_s\right)^2 + \left(\frac{A}{2}\sin\phi_s\right)^2} = \frac{A}{2} \\ \Phi = \text{atan2}(Y, X) = \text{atan2}\left(\frac{A}{2}\sin\phi_s, \frac{A}{2}\cos\phi_s\right) = \phi_s \end{cases}$



- The fused magneto-optical images clearly and completely reveal the characteristic information of defects.
- The BM3D-PSD algorithm effectively integrates useful information from multi-angle raw images.
- The fused image reveals defect morphology that is more complete and clearer than any single original image.
- Blurred or missing defect details in the original images have been effectively reconstructed and enhanced.
- Defect contours in the fused image appear sharp with natural transitions, showing no noticeable blurring or distortion.

Optical Imaging



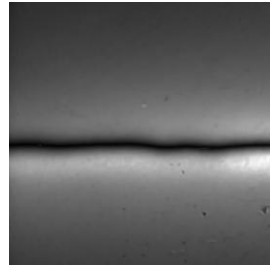
crack



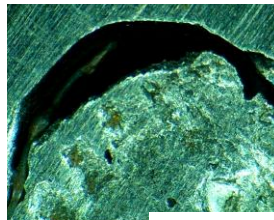
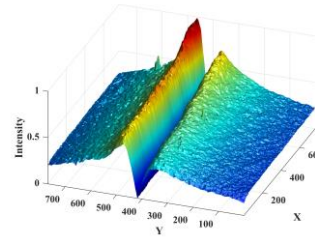
X-ray images



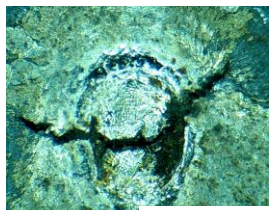
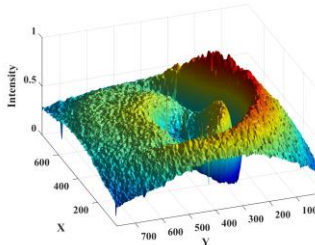
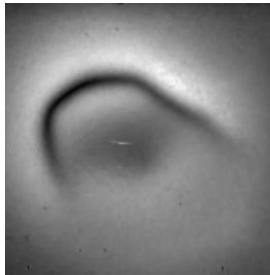
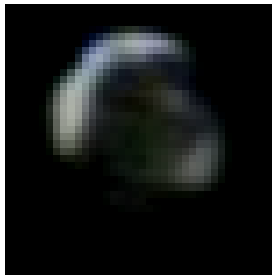
Magneto-optical fusion image



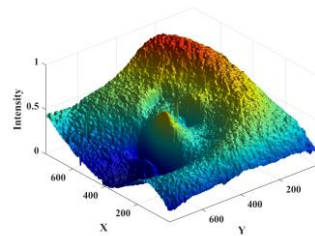
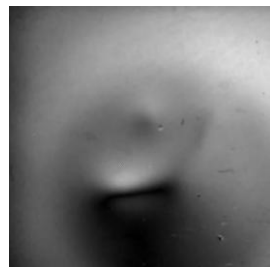
Grayscale 3D



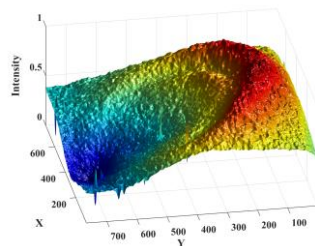
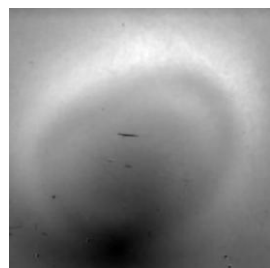
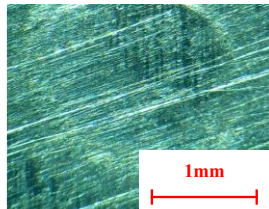
pits



crack



molten core



- Optical images provide surface information but struggle to reveal internal structures.
- X-ray images are sensitive to density variations yet suffer from low resolution and blurred edges.
- Significantly enhances crack edge contrast, displaying internal structures without damaging the sample.
- Grayscale images enable further exploration of defect depth and distribution characteristics, revealing spatial structures for subsequent quantitative assessment.
- Gray-scale images can further investigate defect depth and distribution characteristics, revealing their spatial structure to enable subsequent quantitative evaluation.



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Thank you !